Does Summer Heat Affect Food Affordability in NYC?

Environmental Issue

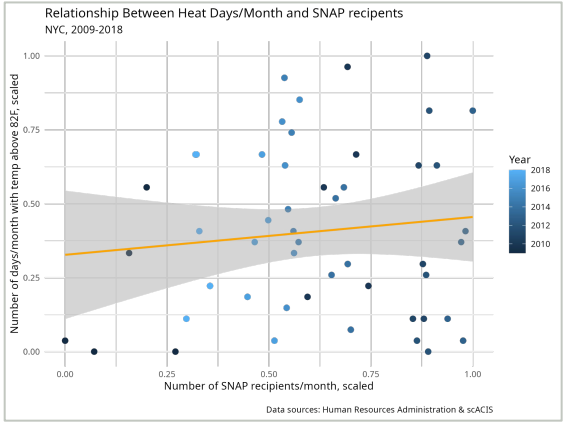
- More hot days → higher cost of living
- Potential link to food insecurity

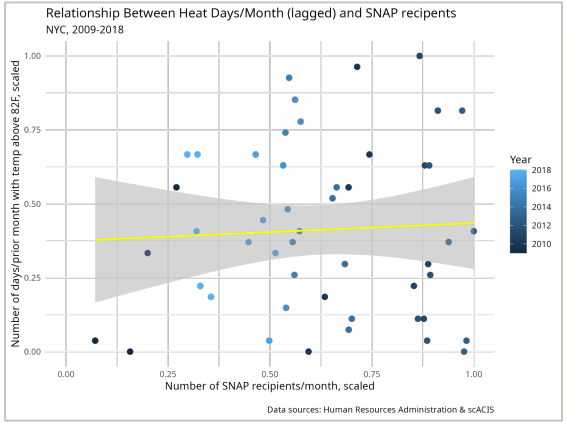
Data & Methods

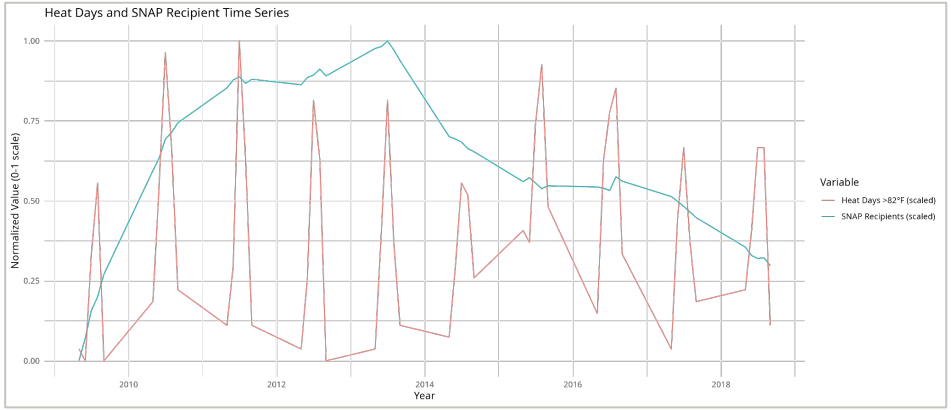
- Monthly SNAP Recipients (2009–2018)
- Heat: days > 82°F (May–Sept)
- Simple regression: heat + SNAP recipients
- Multiple regression: heat + lagged heat + SNAP recipients

Key Finding

SNAP participation is likely driven by long-term economic trends—not short-term heat variation.







Model 1 (simple heat + SNAP)

Heat effect is small and not significant

- Coefficient ≈ 0.10, p = 0.435

Model explains almost nothing

- R^2 ≈ 1.3%

Model 2 (adds lag + time control)

Heat (current & lagged) still not significant

- Current heat: coefficient ≈ 0.26, p = 0.28
- Lagged heat: coefficient ≈ 0.21, p = 0.39

Slightly better apparent fit, but still weak

- R^2 ≈ 15%, adjusted R^2 ≈ ~0%
- Overall model not significant (p = 0.41)

Analysis

While SNAP participation cannot necessarily be explained by variability in heat, there may be other variables that represent food insecurity and affordability that would be worth examining within a similar framework.

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